# ENG3042 Turbojet Lab Report

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**Abstract**

In this lab, we investigated the noise generated from a Cussons P9010e turbojet. This was done to gain an understanding about the sources of the noise in this particular turbojet. The lab showed that the propulsive jet is the main source of noise and this increases as the thrust increases. A thrust level of 148N, this produced a root-mean-square value of 120.4dB which is less than that produced at a thrust of 192N, which produced a value of 123.2dB.

## 1 Introduction

A turbojet is an engine that works by converting low pressure air from outside to hot high pressure exhaust gases which produces thrust. Air is drawn into the rotating compressor from an intake, from there it is compressed to a higher pressure and is then guided into the combustion chamber. The fuel is mixed with high pressure compressed air and burned. The products created are of a high pressure and temperature which are then diverted into the turbine of the turbojet. In the turbine the products are expanded to extract power and used to drive the next stage which is the compressor. The exhaust gases then flow through the exit nozzle and are accelerated to the rear of the turbojet, this is what produces the thrust for the engine. [1]

The noise produced is either mechanical or aerodynamic. The turbojet's main source of noise is the propulsive jet due to aerodynamics. A shear layer causes sound waves to be emitted from the jet. This is due to the pressure changes, that vary with time, caused by the exhaust gases exiting the propelling nozzle. The compressors rotating blades cause a change in pressure which results in sound waves being emitted, but not as loud as that of the propulsive jet. Most components of the turbojet produce noise such as: the compressor, combustion chamber, turbine and the shaft connecting the compressor to the turbine. [2]

We can represent sound waves by using sine waves. Continuous sound waves can be decomposed into their component harmonic sine waves using Fourier analysis. To transform discrete signals, we use the Discrete Fourier Transform. This method is used to analyse sound signals such as voice and music. Fast Fourier Transform (FFT) is the algorithm used to perform this analysis. [3]

## 2 Methodology

The sensors on the turbojet recorded the thrust, air mass flow rate and fuel mass flow rate. To measure the audio frequencies emitted by the turbojet a scientific microphone was used, with built in pressure sensors, and connected to a computer to record the data. The data acquisition system sampled 3 channels at a rate of 40000 samples per second per channel for one second. The data was collected for a thrust of 148N and 192N and saved as dat files. Using MATLAB and Fast Fourier Transform (FFT) to analyse the square wave signal the data analysis methodology can be verified (Figure 1).

Maximum sound pressure level (SPL) magnitude and the root-mean-square can be found for the 2 signals by converting the microphone signal into pressure, Pa. These values are then converted to decibels (Table 1). The Fourier mode amplitudes were plotted as a function of frequency by using Fourier analysis on the microphone signal (Figure 2a and 2b).

## 3 Results and Discussion

In Figure 1 we can see that the FFT of the square wave signal is a function of frequency. The first peak is 262Hz. This is 1Hz more than the fundamental frequency given. The second peak is 3(262Hz) and the third peak is 5(262Hz). Figure 1 verifies that the data analysis methodology is correct, considering experimental errors.

It can be seen in Table 1 that as the thrust is increased, the noise increases as rms values increase from 120dB at 148N to 123dB at 192N of thrust. This can be put down to a greater amount of fuel and air through the system which in turn causes the compressor and turbine to rotate much faster, which means there will be a larger explosion in the combustion chamber and therefore more exhaust gases exiting the propelling nozzle at a much higher jet velocity. A greater change in pressure is created by a higher jet velocity which in turn creates a greater shear layer.

The FFT of the microphone signal as a function of frequency is shown in Figure 2a and 2b. The harmonics of rotational frequency are shown by the sequences of peaks in Figure 2a. They are as follows: 6547Hz, 8195Hz and 9841Hz. The Blade Passing Frequency (BPF) is 11478Hz. To calculate the RPM of the turbojet engine, use the following method: The difference between the BPF and neighbouring peaks and multiply by 60.

See Table 1 for RPM values at thrust levels 148N and 192N. These values were calculated using method mentioned previously. The lower frequency section ranges from a very low value to a few kHz and the peak occurs at 512Hz; this is due to jet noise. In Figure 2b the amplitude increases in the lower frequency section. This further proves that the sound level of the turbojet increases as the thrust increases.

## 4 Conclusions

From this lab it was found that as the sound level of the turbojet increases as the thrust increases. This is because there is a greater shear layer because of an increased jet velocity. A greater shear layer results in a greater pressure change and therefore produces greater sound waves. The sound level of the turbojet increased when the thrust increased as compressor blades rotate faster which caused a greater change in pressure.

## 5 References

[1] McMahon, P.J 1971, Aircraft Propulsion, Pitman, London, Pg 80-82

[2] Archer, R.D & Saarlas, M. 1996, Introduction to aerospace propulsion, Prentice Hall, London; Upper Saddle River, NJ;. Pg 496

[3] Huggins, E., 2000. Physics 2000. [ebook] Chapter 16 <https://physics2000.com/PDF/Text/Ch_16_FOURIER_ANALYSIS,_NORMAL_MODES_AND_SOUND.pdf>

## 6 Tables

| Thrust (N) | Mass Flow Rate (kg/s) | Jet Velocity (m/s) | SPL (dB) | RMS (dB) | RPM |
|---|---|---|---|---|---|
| 148 | 0.415 | 357 | 134 | 120 | 98400 |
| 192 | 0.444 | 432 | 135 | 123 | 105500 |

Table 1: Test Data for Turbojet

## 7 Figures

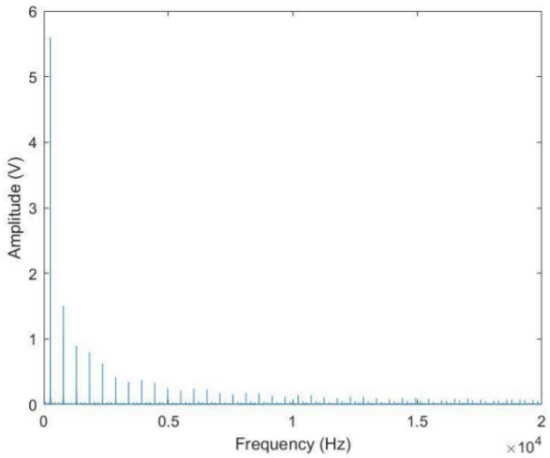



Figure 1: Amplitude Spectrum of Square Wave Signal

Figure 2a: Thrust 148N

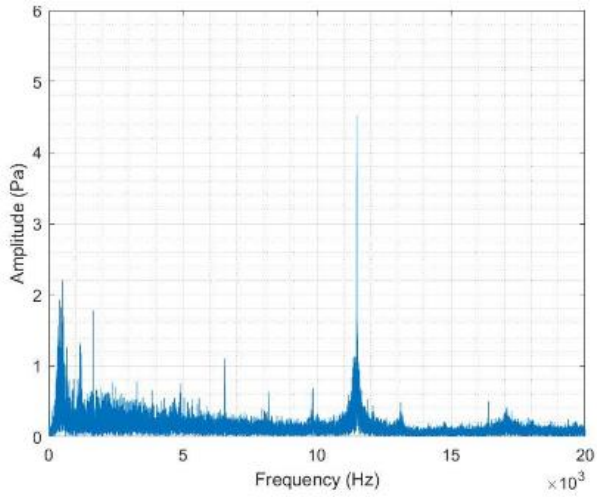



Figure 2b: Thrust 192N

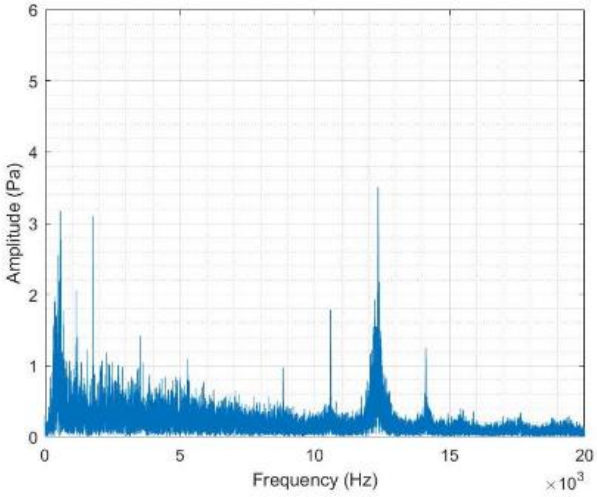



Figure 2: Amplitude Spectrum of Microphone Pressure Signal